

Random-shape fix validation — 2D

Both original hull rejections fixed without changing core geometry.

Both original concave-drop references now qualify.

One of two additional seeds qualifies; one still fails.

Both packed 36-body trajectories remain unqualified.

Controller: 1.37–2.46× lower native time than its continuous fine setting.

It is slower than the cheapest passing fixed setting in all three cases.

Offline selector returns no accuracy-backed choice for failed references.

60 retained histories; unchanged budgets; independent geometry/source audit.

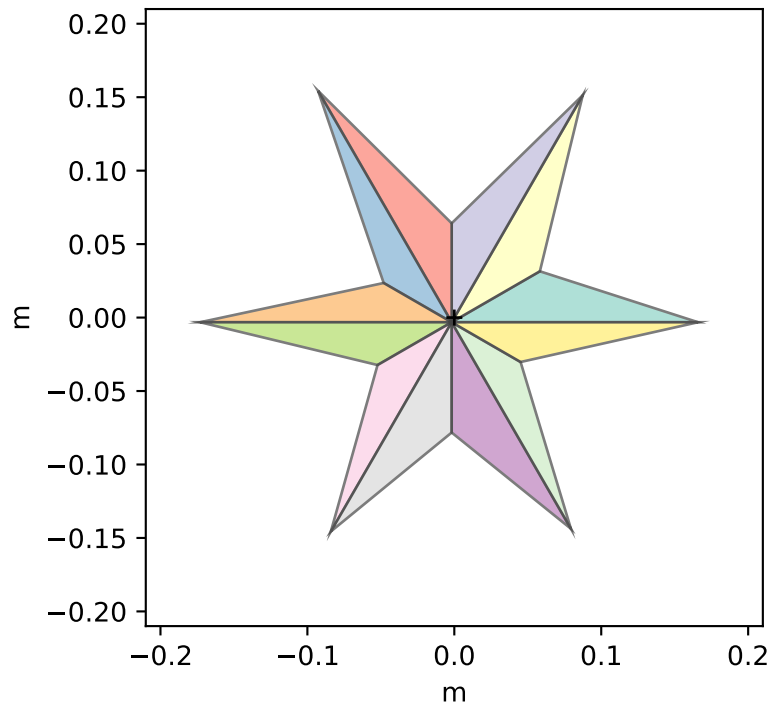
Full Float64, analytic wall motion and exact partition controls tested.

86 local tests pass. No 3D, material validation or universal engine claim.

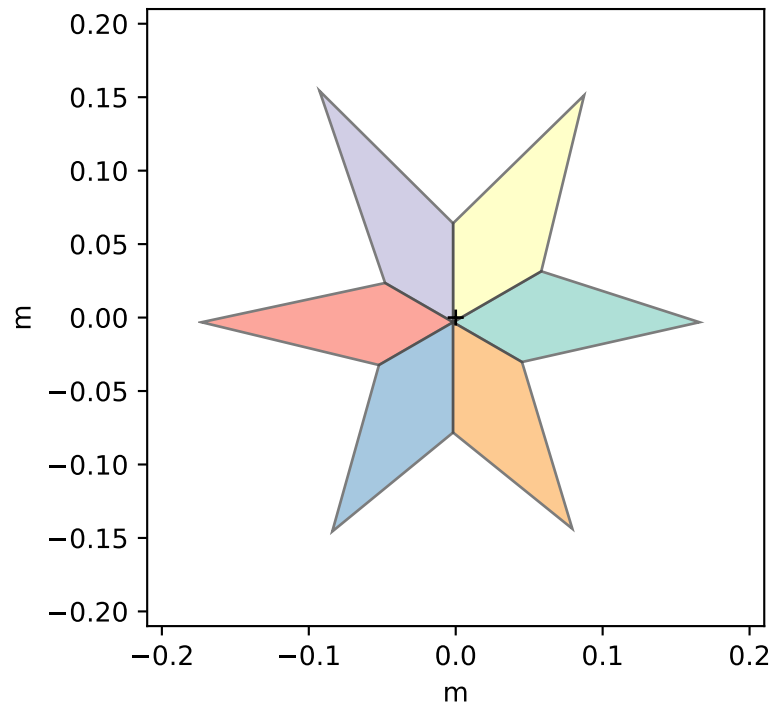
Complete tables, source pins and remaining work: [report.md](#) and [results/](#).

Exact representation change: no boundary smoothing; mass and inertia preserved

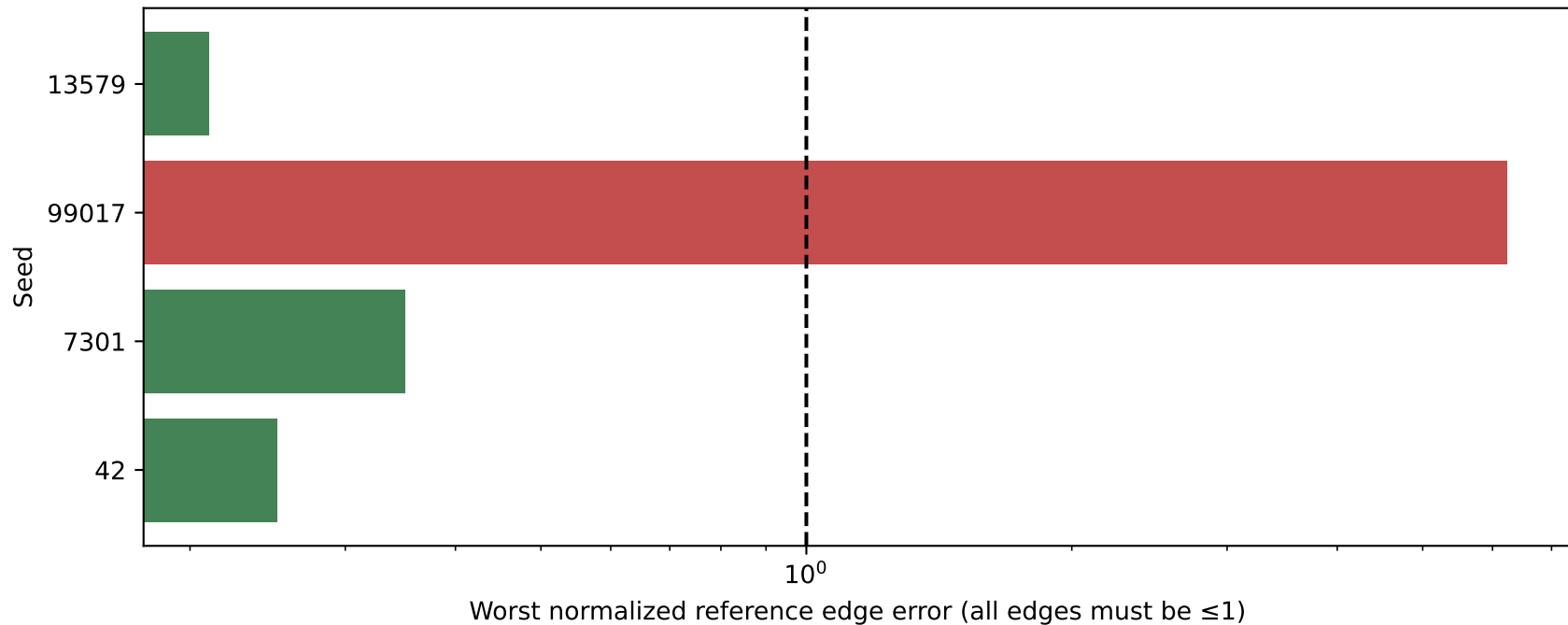
Original 12 triangle cores



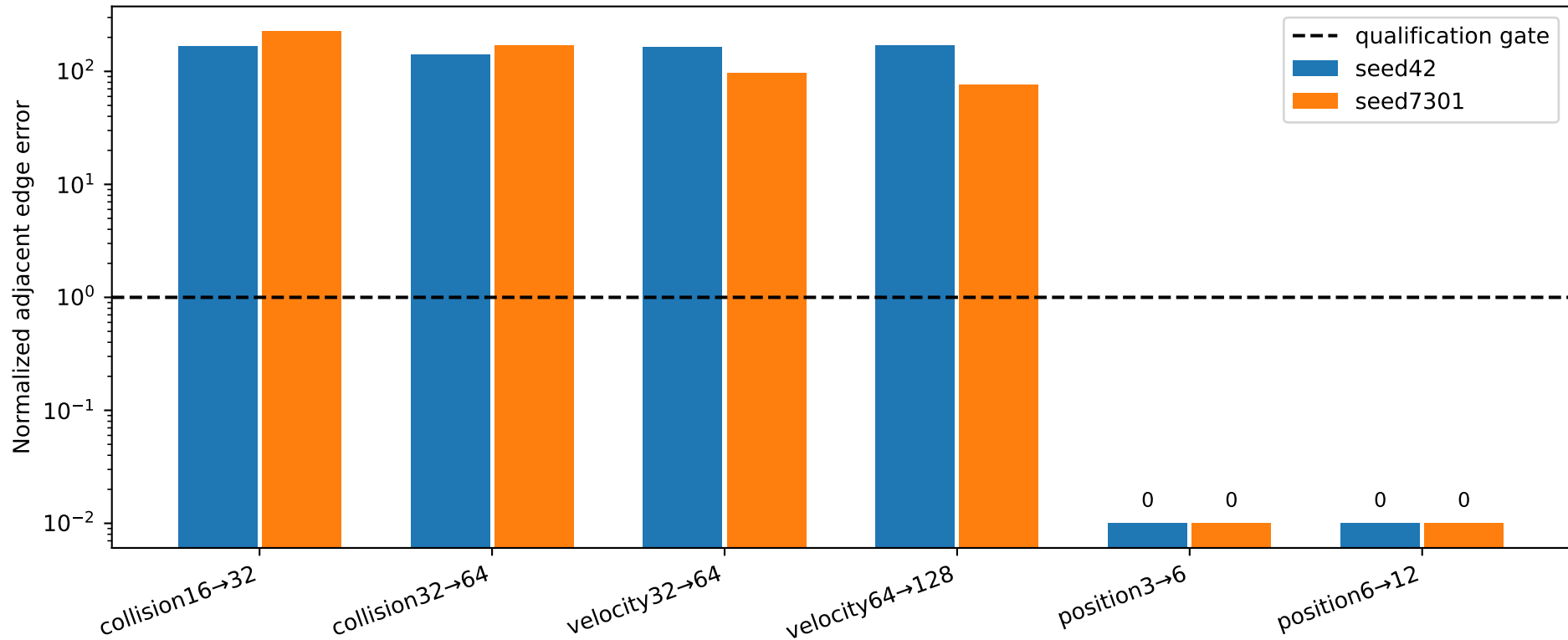
Same union, 6 convex cores



Original cases qualify; held-out seed99017 remains unresolved



Packed boxes: Float64 + exact partition + analytic walls still fail



Controller saves idle work; fixed settings remain cheaper

